

SAMSUN CARSAMBA, Turkey

WMO: 170310

Lat: 41.258N

Lon: 36.556E

Elev: 7

StdP: 101.24

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.0	-8.2	1.9	3.7	-6.1	2.3	4.3	13.2	3.9	11.7	4.7	2.1	130	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.5	30.2	22.7	29.0	22.3	28.1	22.1	24.5	28.2	23.8	27.3	23.2	26.8	3.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	17.9	27.0	22.6	17.3	26.5	21.9	16.6	26.0	74.3	28.3	71.4	27.5	68.9	26.8	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.6	7.3	-5.8	33.7	2.8	1.6	-7.8	34.8	-9.4	35.7	-11.0	36.6	-13.0	37.7		
			-6.4	25.3	2.6	0.9	-8.3	25.9	-9.8	26.4	-11.2	26.9	-13.1	27.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	6.8	7.4	8.9	11.6	16.2	20.6	23.0	23.7	20.2	16.2	11.8	8.2	(6)
		DBStd	6.74	4.04	4.11	3.78	3.40	2.86	2.23	1.70	1.75	2.32	2.99	3.41	3.73	
		HDD10.0	393	116	92	68	18	0	0	0	0	1	19	79		
		HDD18.3	1844	357	306	294	203	77	6	0	10	80	197	314		
		CDD10.0	2069	17	20	33	66	194	317	404	424	304	191	73	24	
		CDD18.3	478	0	0	1	2	12	73	146	166	64	12	1	1	
		CDH23.3	3434	0	3	16	42	92	436	1067	1304	364	82	24	3	
		CDH26.7	627	0	0	3	11	17	73	187	264	52	18	2	0	
(14)	Wind	WSAvg	2.7	3.1	3.1	2.9	2.6	2.3	2.6	2.9	2.9	2.5	2.4	2.6	2.9	(14)
(15)	Precipitation	PrecAvg	874	87	62	76	50	52	53	40	49	78	110	104	101	(15)
(16)		PrecMax	1187	124	109	146	76	186	111	121	179	170	228	255	167	(16)
(17)		PrecMin	586	11	26	32	18	12	6	1	5	27	49	12	53	(17)
(18)		PrecStd	124	30	25	28	19	40	28	32	45	36	45	66	33	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	20.2	23.0	25.7	28.0	28.4	31.1	31.1	31.8	30.2	29.0	26.5	22.9		(19)
(20)		MCWB	10.5	12.2	13.6	16.3	19.3	21.5	23.3	23.7	21.1	18.6	14.9	12.7		(20)
(21)		DB	18.1	19.9	21.8	23.8	25.9	28.2	29.8	30.1	28.0	25.5	23.1	19.7		(21)
(22)		MCWB	9.7	11.0	12.2	14.9	18.1	21.1	22.8	23.8	21.5	18.3	14.2	11.4		(22)
(23)		DB	16.0	17.1	18.8	20.8	23.8	27.0	28.7	29.0	26.6	23.1	20.8	17.1		(23)
(24)		MCWB	8.9	9.9	11.2	13.7	17.7	20.9	22.5	23.2	21.0	18.1	13.5	10.3		(24)
(25)		DB	14.0	14.8	15.7	17.9	21.9	25.9	27.8	28.1	25.3	21.2	18.0	14.8		(25)
(26)		MCWB	8.0	8.9	10.2	12.8	17.0	20.3	22.3	22.8	20.4	17.3	12.6	9.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	11.3	13.1	14.2	17.7	20.5	23.6	25.2	26.0	23.6	20.6	17.4	13.7		(27)
(28)		MCDB	18.3	20.1	23.3	25.6	26.1	27.4	28.9	30.0	27.1	25.1	21.0	19.3		(28)
(29)		WB	10.2	11.5	12.7	15.5	19.3	22.7	24.3	24.8	22.4	19.5	15.6	12.3		(29)
(30)		MCDB	16.2	18.3	19.8	22.2	24.2	26.4	27.9	28.7	26.2	23.3	19.9	17.4		(30)
(31)		WB	9.5	10.4	11.6	14.2	18.4	21.8	23.6	24.1	21.6	18.6	14.5	11.2		(31)
(32)		MCDB	14.3	16.0	17.7	19.5	22.5	25.7	27.2	27.7	25.4	22.1	18.7	15.7		(32)
(33)		WB	8.6	9.3	10.4	13.0	17.5	21.0	23.0	23.5	20.8	17.9	13.5	10.3		(33)
(34)		MCDB	12.4	13.7	14.9	17.3	20.9	24.7	26.5	27.0	24.5	21.0	17.5	13.8		(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	8.8	9.1	9.7	9.4	9.2	8.5	8.5	9.4	9.0	10.4	9.2		(35)
(36)		5% DB	12.3	14.3	15.9	16.3	13.3	11.7	10.4	9.9	11.6	13.0	15.4	13.7		(36)
(37)		MCWBR	6.1	6.9	7.5	8.1	6.8	6.0	5.3	5.1	6.1	7.1	7.8	7.2		(37)
(38)		5% WB	10.7	12.8	14.3	14.3	11.0	9.7	8.4	8.6	10.2	10.9	12.9	11.6		(38)
(39)		MCWBR	5.9	6.7	7.2	7.6	6.1	5.4	5.0	5.0	5.8	6.4	7.4	6.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.352	0.387	0.431	0.473	0.444	0.419	0.435	0.445	0.414	0.406	0.365	0.352		(40)
(41)		taud	2.333	2.229	2.131	2.055	2.222	2.338	2.318	2.278	2.357	2.319	2.384	2.361		(41)
(42)		Ebn at Noon	782	804	809	802	837	858	840	819	821	777	763	749		(42)
(43)		Edh at Noon	88	113	139	161	140	125	126	128	110	103	84	80		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.62	2.35	3.38	4.69	5.73	6.60	6.51	5.73	4.44	2.94	2.03	1.44		(44)
(45)		RadStd	0.18	0.29	0.27	0.48	0.57	0.52	0.34	0.33	0.31	0.26	0.23	0.18		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-154	N/A	N/A
Regional (6 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	-82	-143	+106	+66

Nomenclature: See separate page